Most General Winning Secure Equilibria Synthesis

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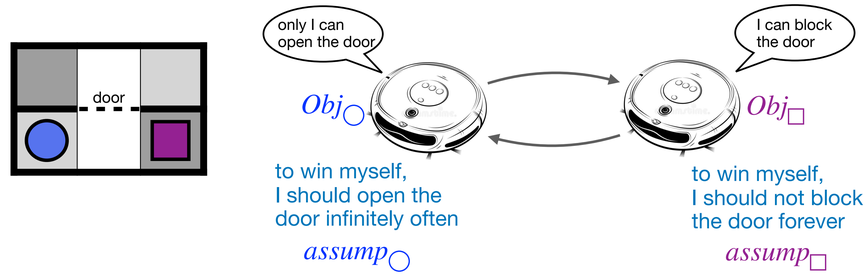
Rational Robots in a Workspace

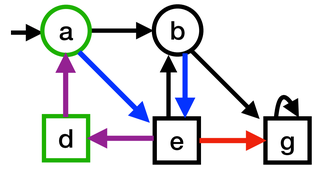


Secure equilibrium (SE) = cooperative strategy + punishment strategy

alternately use the middle passage — block the door forever

How to generalize secure equilibria to have more flexibility for the systems?

Rational Players in a Graph Game



○ vs □

$Obj_{\bigcirc}$ = infinitely often **a**

$Goal_{\bigcirc} = \left(Obj_{\bigcirc}, \neg Obj_{\square}\right)$

$Obj_{\square}$ = infinitely often **d**

$Goal_{\square} = \left(Obj_{\square}, \neg Obj_{\bigcirc}\right)$

Winning Secure equilibrium (WSE)

$(Str_{\bigcirc}, Str_{\square})$

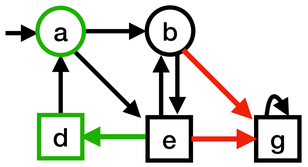
- $(Str_{\bigcirc}, Str_{\square}) \models$ both win
- $Str_{\bigcirc} \models$ ○ loses ⇒ □ also loses
- $Str_{\square} \models$ □ loses ⇒ ○ also loses

cooperative strategy + punishment strategy

$Str^c_{\bigcirc}$: **..a → e**, **..b → e** + $Str^p_{\bigcirc}$

$Str^c_{\square}$: **..e → d**, **..d → a** + $Str^p_{\square}$: **..e → g**, **..d → a**

Most General Winning Secure Equilibria



○ vs □

$Obj_{\bigcirc}$ = infinitely often **a**

↓

$assump_{\bigcirc}$ = never (**b** → **g**)

$Obj_{\square}$ = infinitely often **d**

↓

$assump_{\square}$ = never (**e** → **g**)

∧ inf (**e**) ⇒ inf (**e** → **d**)

Most General WSE

$\left(\Psi_{\bigcirc}, \Psi_{\square}\right)$

- $\Psi_{\bigcirc} \wedge \Psi_{\square} \equiv Obj_{\bigcirc} \wedge Obj_{\square}$
- each Ψ_i is realizable by Player i
- every $(Str_{\bigcirc}, Str_{\square})$ with $Str_i \models \Psi_i$ forms a **WSE**

$$\Psi_{\bigcirc} = assump_{\bigcirc} \wedge (assump_{\square} \Rightarrow Obj_{\bigcirc}) = assump_{\bigcirc}$$

$$\Psi_{\square} = assump_{\square} \wedge (assump_{\bigcirc} \Rightarrow Obj_{\square}) = assump_{\square}$$

Contribution

- most general WSE = collection of equilibria as **independently** realizable specifications
- **sound** and **efficient** but incomplete algorithm
- generalized to k-player games (even with **Env**)

Future Works

- extend the notion to other equilibria, e.g., subgame-perfect equilibria
- quantitative settings